Kushal Kejriwal

kkejriwa@andrew.cmu.edu | |

EDUCATION

Carnegie Mellon University — Pittsburgh, Pennsylvania, USA
Master of Science in Computer Science. GPA: 4.0/4.0. — Aug 2026 – Present

Indian Institute of Technology Bombay — Mumbai, India
Bachelor + Master of Technology, **Major**: Electrical Engineering; **Minor**: Machine Learning — Aug 2019 – Jun 2024
Department Rank: 1, Institute Silver Medal. GPA: 9.80/10.0
Teaching Assistant: Calculus, Image Processing, Online Learning and Optimization

EXPERIENCE

Jane Street — May 2022 – Jul 2022
Quantitative Trader Intern — Hong Kong

- Built rolling linear regression models to predict returns and designed data-driven trading strategies across markets
- Backtested strategies on four years of historical data, achieving statistically significant risk-adjusted returns (t-stat > 2)

Optiver — Aug 2024 – Feb 2026
Quantitative Trader — Amsterdam, Netherlands

- Independently traded and managed a portfolio of 14 European single-stock options across two sectors, actively managing risk
- Developed regression models to forecast NASDAQ closing auction imbalances, improving strategy PnL by 17%
- Built a Dash application in Python to display trading metrics, adopted in production and used to inform trading decisions

RESEARCH

Training RNNs with PPO | CMU — Aug 2026 – Present
Reinforcement Learning, Policy Optimization, Recurrent Neural Networks, Distributed Training

- Formulating a PPO (Proximal Policy Optimization) framework for training RNNs (Recurrent Neural Networks) to improve reinforcement learning post-training stability for longer context lengths
- Benchmarking against related reinforcement learning fine-tuning techniques such as GRPO to evaluate credit assignment quality and training stability at scale
- Leveraging Google Cloud TPUs to scale model training, porting the framework to TPU-based infrastructure for experiments

Decentralized Heterogeneous LoRA Fine-Tuning | MBZUAI — Mar 2026 – Aug 2026
Large Language Models, Parameter-Efficient Fine-Tuning, Federated Learning, Post-Training — Abu Dhabi, UAE

- Proposed a decentralized LoRA algorithm for supervised fine-tuning of language models in LLM post-training, enabling structured collaboration between heterogeneous clients
- Implemented the framework in PyTorch and trained Qwen 3B language models across 2 NVIDIA A6000 GPUs
- Conducted empirical evaluations across 30+ tasks, achieving approximately 2% ROUGE-L improvement over state-of-the-art federated learning baselines

Fixed-Confidence Multi-Armed Bandits | IIT Bombay — Jul 2023 – Jul 2024
Multi-Armed Bandits, Stochastic Optimization, Sequential Decision Making, Algorithms

- Studied identification of the best arm under uncertainty with minimal trials (best-arm identification in multi-armed bandits)
- Proposed LUCB-Greedy, a simplified algorithm approximately 30% faster than baseline algorithms while preserving optimality

PUBLICATIONS

- **K. Kejriwal**, N. Karamchandani, & J. Nair; On the Asymptotic Optimality of Confidence-Interval-Based Algorithms for Fixed-Confidence MABs; **accepted at AAAI 2024**.
- A. Biswas, N. S. Nambiar, **K. Kejriwal** & U. Ganguly; MAdapter: A Multimodal Adapter for Liquid State Machines configures the Input Layer for the same Reservoir to enable Vision and Speech Classification; **accepted at IJCNN 2023**.

SKILLS

Programming Languages: Python, C, C++, Go, R, JavaScript
ML Libraries: PyTorch, TensorFlow
Machine Learning: Deep Learning, Reinforcement Learning, Optimization for Machine Learning, Computer Vision
Applied Mathematics: Advanced Probability and Stochastic Processes, Online Learning, Optimization Techniques, Markov Chains, Queuing Systems, Information Theory, Error Correcting Codes
Computer Science: Distributed Systems, Computer Systems, Digital Systems, Computer Programming and Utilization